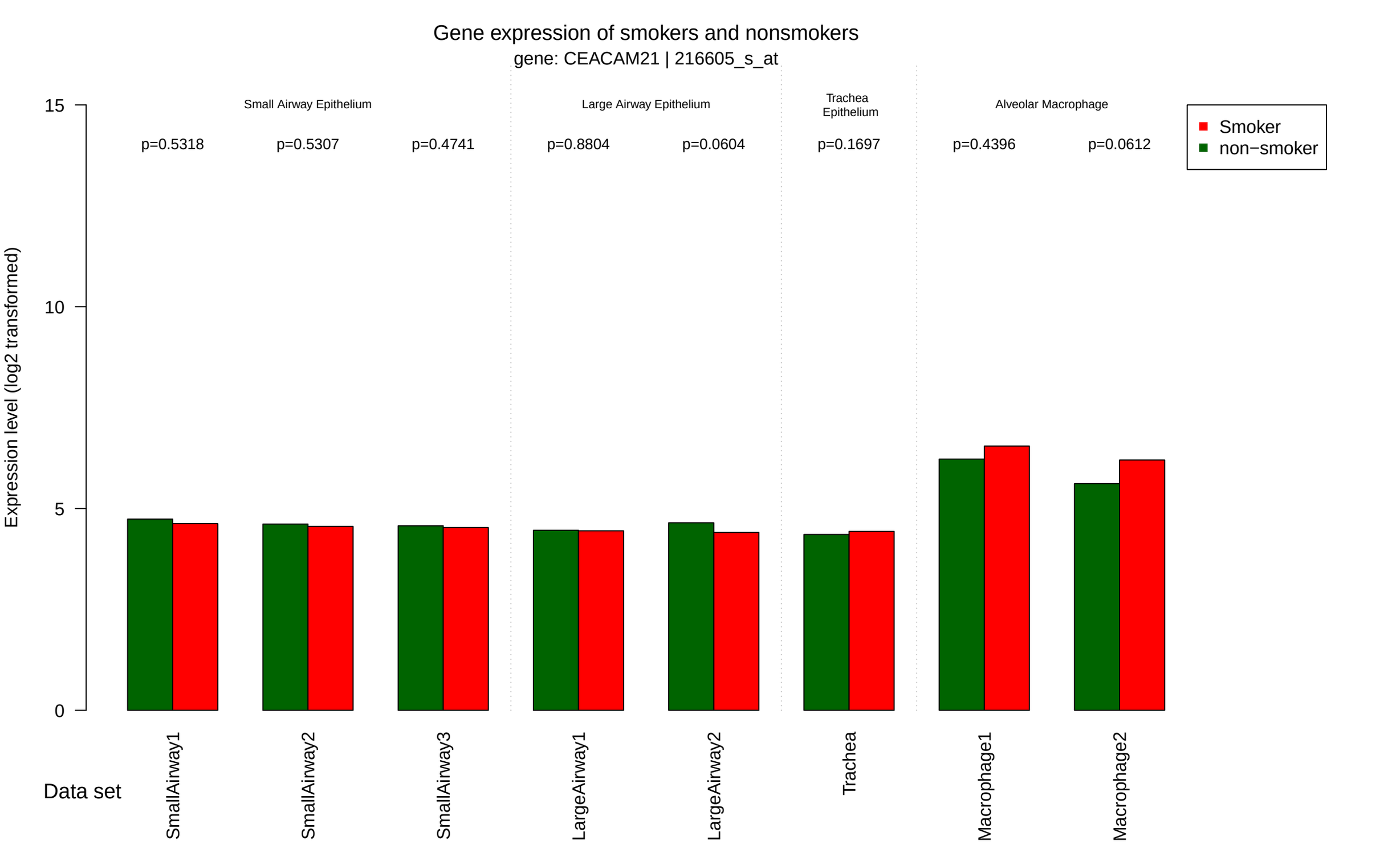
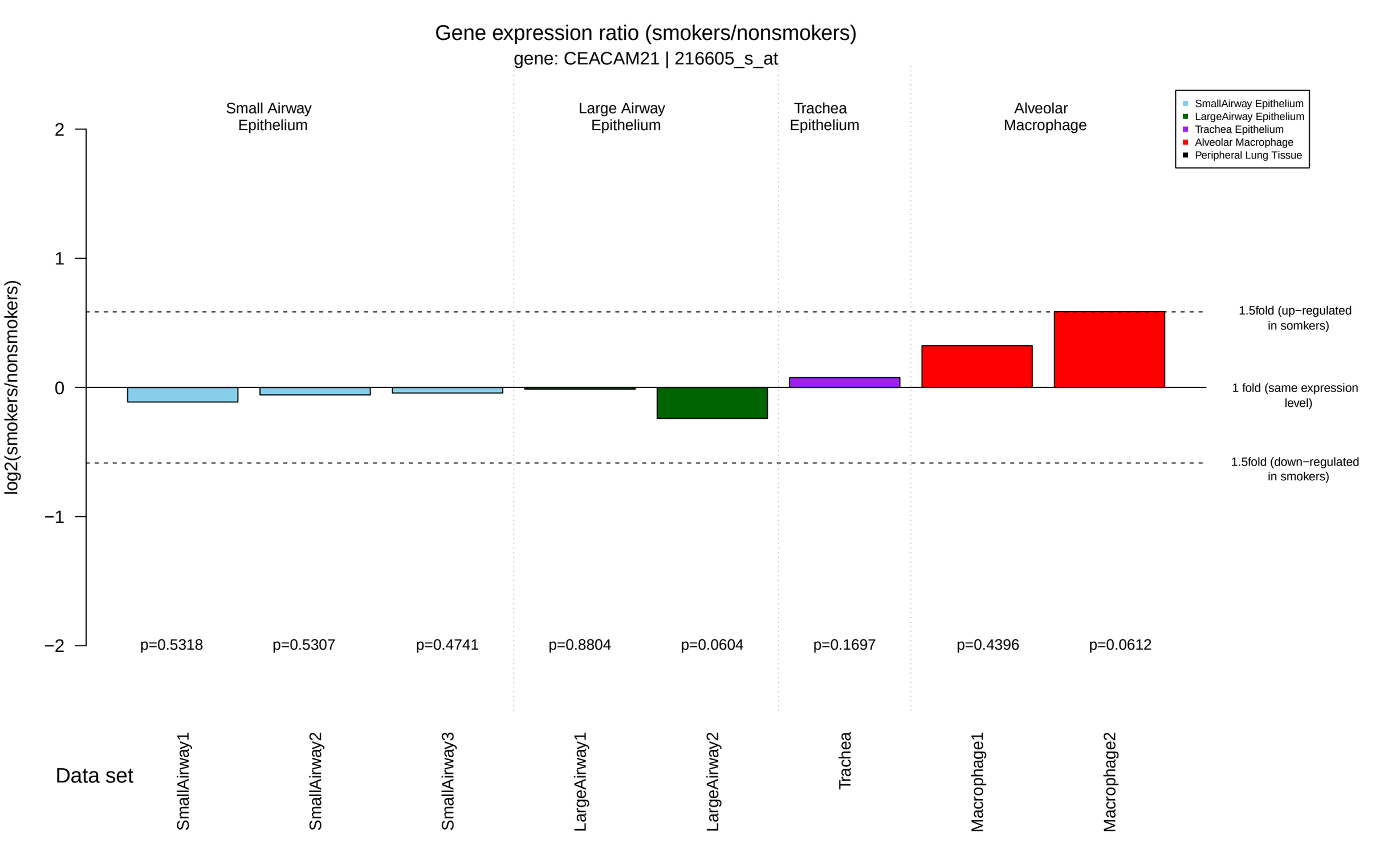
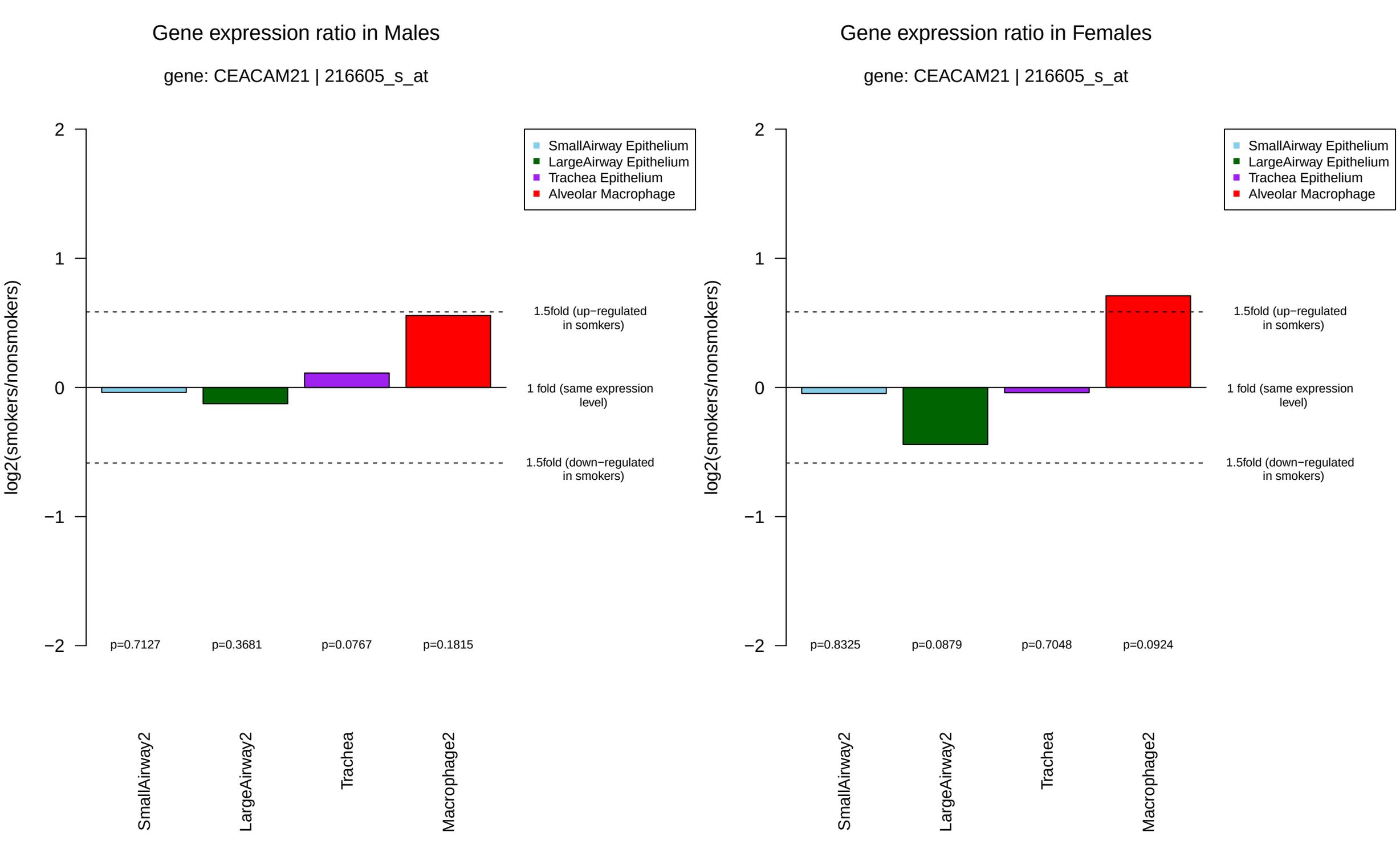
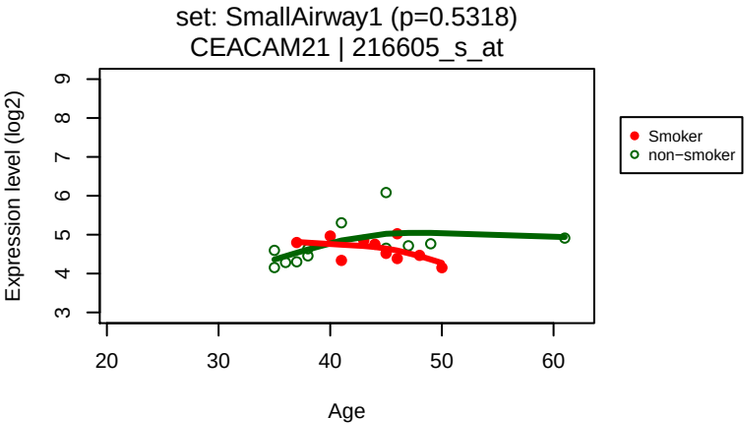
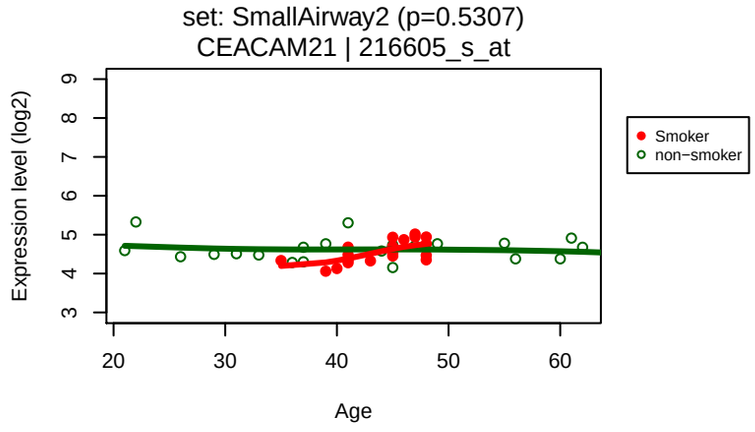
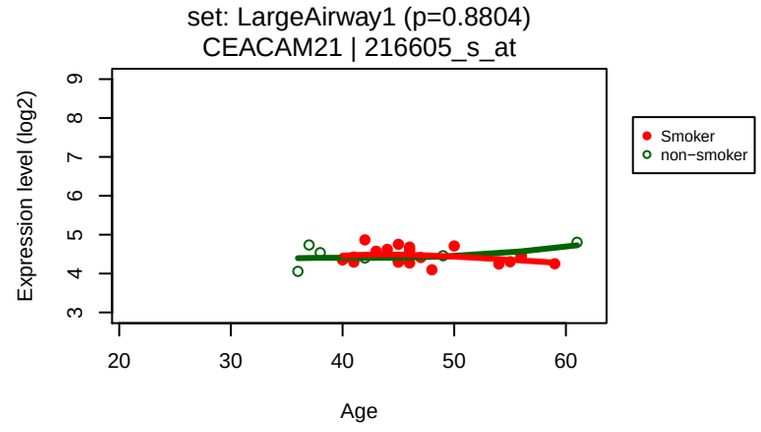

Gene expression of smokers and nonsmokers
gene: CEACAM21 | 216605_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5318
p=0.5307
p=0.4741
p=0.8804
p=0.0604
p=0.1697
p=0.4396
p=0.0612
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

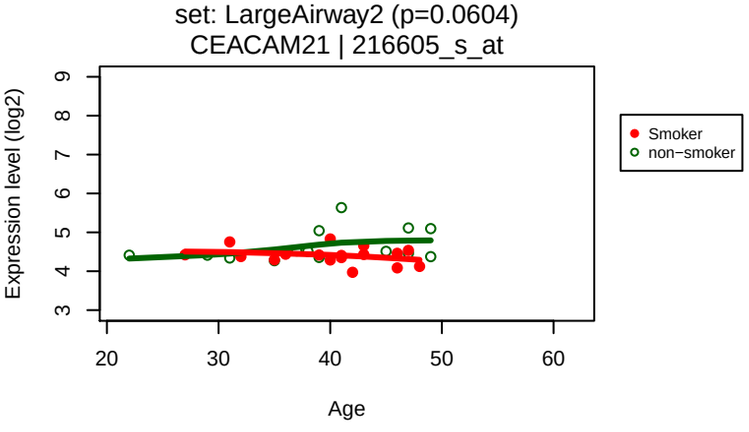

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM21 | 216605_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5318
p=0.5307
p=0.4741
p=0.8804
p=0.0604
p=0.1697
p=0.4396
p=0.0612
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

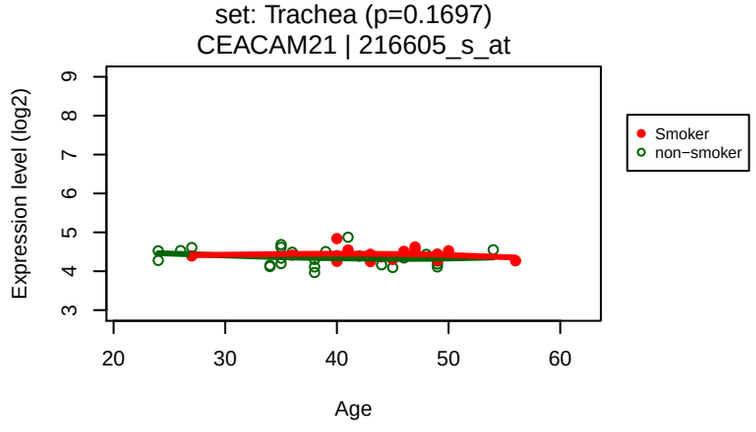

Gene expression ratio in Males
gene: CEACAM21 | 216605_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7127
p=0.3681
p=0.0767
p=0.1815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM21 | 216605_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8325
p=0.0879
p=0.7048
p=0.0924
SmallAirway2
LargeAirway2
Trachea
Macrophage2

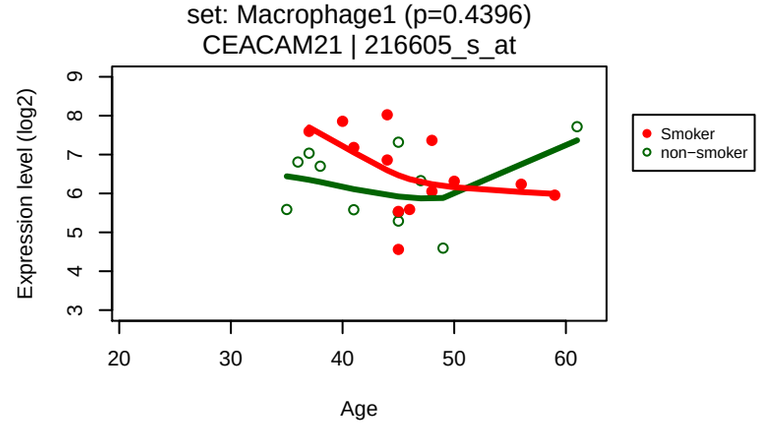

set: SmallAirway1 (p=0.5318)
CEACAM21 | 216605_s_at
Expression level (log2)
Age
Smoker
non−smoker

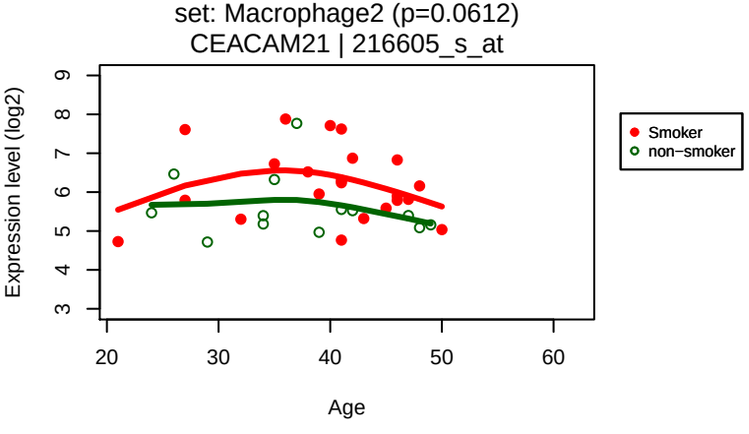

set: SmallAirway2 (p=0.5307)
CEACAM21 | 216605_s_at
Expression level (log2)
Age
Smoker
non−smoker

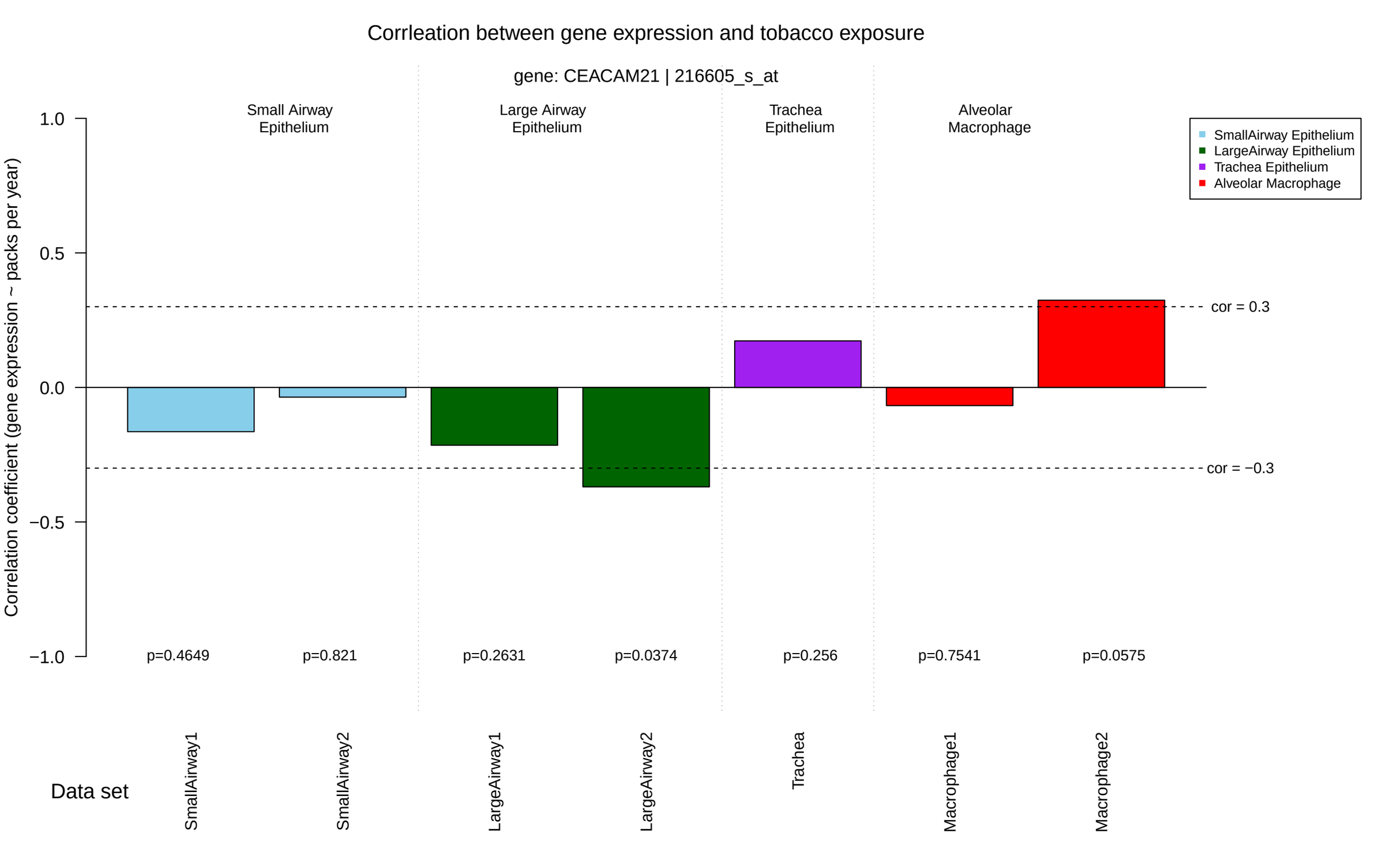

set: LargeAirway1 (p=0.8804)
CEACAM21 | 216605_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0604)
CEACAM21 | 216605_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1697)
CEACAM21 | 216605_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4396)
CEACAM21 | 216605_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0612)
CEACAM21 | 216605_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM21 | 216605_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4649
p=0.821
p=0.2631
p=0.0374
p=0.256
p=0.7541
p=0.0575
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2